STUDENT DETAILS USING ARRAY OF STRUCTURES

AIM:

Write a program to

a) Define a structure Student with members:Roll number, Name, Marks of three subjects
b) Read the details of n students.
c) Compute the total marks of each student.
d) Display all student details (roll no, name, marks, total) in a tabular form.

PROGRAM:

```
#include <stdio.h>

struct Student {
   int rollNo;
   char name[50];
   int marks[3];  // marks of 3 subjects
   int total;     // total marks
};

void main() {
   int n, i, j;
   printf("Enter number of students: ");
   scanf("%d", &n);
   struct Student s[n];
   for (i = 0; i < n; i++) {
      printf("\nEnter details of student %d\n", i + 1);
      printf("Roll No: ");
      scanf("%d", &s[i].rollNo);
      printf("Name: ");
      scanf("%s", s[i].name);

      s[i].total = 0; // initialize total to 0
      for (j = 0; j < 3; j++) {
         printf("Marks of subject %d: ", j + 1);
         scanf("%d", &s[i].marks[j]);
         s[i].total += s[i].marks[j];
      }
   }
   printf("\n\nStudent Details");
```

```
    printf("\nRoll No\t\tName\t\tSub1\t\tSub2\t\tSub3\t\tTotal\n");
    printf("-----------------------------------------------------------------\n");

    for (i = 0; i < n; i++) {
        printf("%d\t\t\t%s\t\t\t%d\t\t\t%d\t\t\t%d\t\t\t%d\n",
                s[i].rollNo, s[i].name,
                s[i].marks[0], s[i].marks[1], s[i].marks[2], s[i].total);
     }
  }
```

ADDITION OF COMPLEX NUMBERS USING STRUCTURES

AIM:

Write a program using structure to add two complex numbers stored in two structures and store the result in a third structure. Members of structure are real, imaginary parts of a complex number. (Complex number is of the form a+ib, where a-real part, b-imaginary part).

PROGRAM:

```
#include <stdio.h>

struct Complex {
    float real;
    float imag;
};

void main() {
    struct Complex c1, c2, sum;
    printf("Enter real and imaginary part of first complex number: ");
    scanf("%f %f", &c1.real, &c1.imag);

    printf("Enter real and imaginary part of second complex number: ");
    scanf("%f %f", &c2.real, &c2.imag);

    sum.real = c1.real + c2.real;
    sum.imag = c1.imag + c2.imag;

   printf("\nFirst Complex Number: %.2f + %.2fi", c1.real, c1.imag);
    printf("\nSecond Complex Number: %.2f + %.2fi", c2.real, c2.imag);
```

```
    printf("\nSum of Complex Numbers: %.2f + %.2fi\n", sum.real, sum.imag);

}
```